Lecture 8: On "Preferences" and Preference Data

RLHFBOOK.COM

Nathan Lambert

Post-training course.
Chapters 10 & 11.

How do we hillclimb on capabilities we don’t really know how to score?

Is it easier to write a good demonstration or decide between a good and a bad example?

Which poem is better? Which model?

Same prompt to two frontier models (Claude 3.7 Sonnet and ChatGPT with GPT-4o, Feb 2025): *"Write me a short poem about an optimistic goldfish."* Which is better – and which came from which?

ASSISTANT (A)

The Optimistic Goldfish

A little goldfish, bright and bold,
Swam in circles, dreams untold.
Though the bowl was small and round,
Endless wonders still abound.

"The world is vast," she'd always say,
"I'll find new joys in every day!"
No walls could dim her hopeful view,
For in her heart, the ocean grew.

ASSISTANT (B)

The Optimistic Goldfish

In a bowl of curved glass,
Swims a goldfish, bright and bold.
Three seconds of memory, they say,
Yet each circuit feels brand new and gold.

"Today might bring fresh flakes,
Or perhaps a castle tall.
What adventure awaits beyond this wall?"
The optimist in golden hue.

When preference replaces correctness

Most of post-training so far (SFT, RLVR) has had a notion of *correct* – a math answer, a passing test, a verifiable reward. Preferences are what we reach for when there is **no right answer**.

E.g. Which poem is better? Which explanation is clearer? Which tone is kinder?

Preferences came to be the core of RLHF, as the alignment community looked to optimize for complicated objectives like "human values."

This lecture

We trace where “preferences” came from, argue why it’s an **imperfect problem formulation**, then dig into **preference data** (chapter 11) used for today’s models.

The plan

1. A short **history** of preferences – and why it’s an **imperfect** formulation (chapter 10)
2. **Preference data** – the trade-offs in practice (chapter 11)
3. **Open questions** (some shared with synthetic data)

The paper behind the first half of this lecture

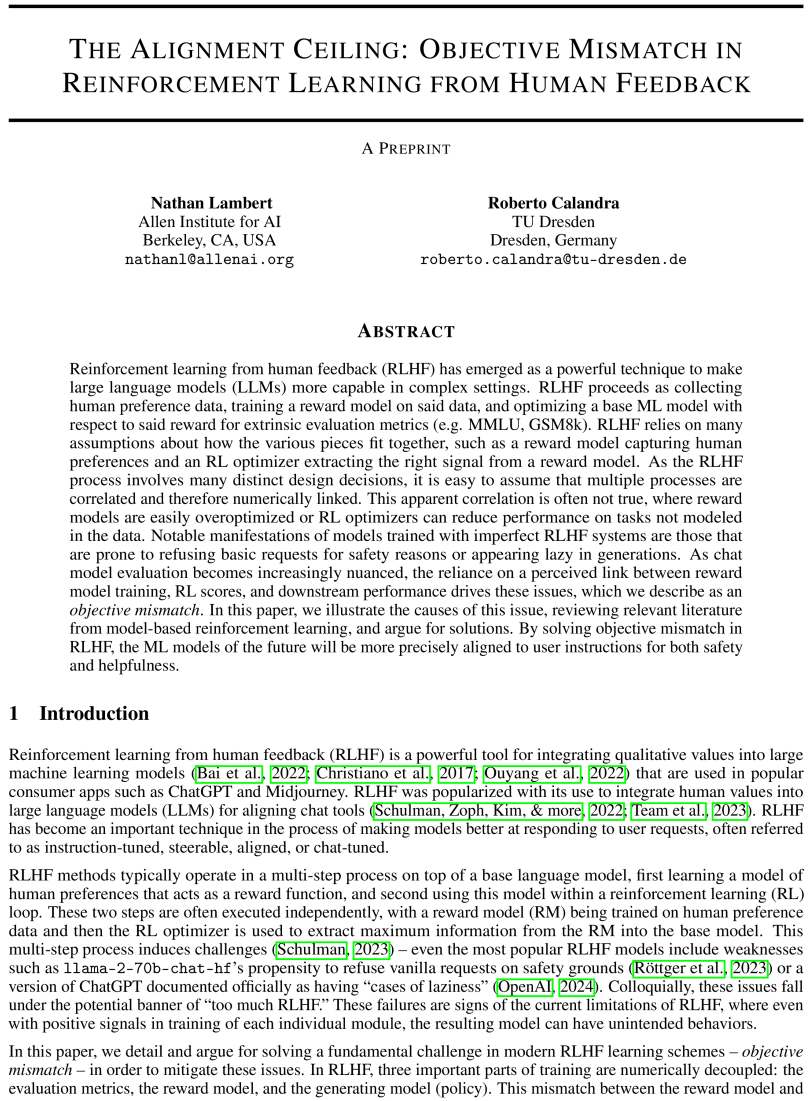
Lambert et al., 2023

Aside: The "objective mismatch" lens

A recurring way to think about post-training: we optimize a **proxy** objective that isn't the one we actually care about.

- **Model-based RL:** the dynamics model is trained for prediction accuracy, not control performance – so a "better" model can yield a *worse* policy (Lambert et al., 2020).
- **RLHF:** the reward model is trained for preference-classification accuracy, not downstream policy quality – the same mismatch, one level up (Lambert & Calandra, 2023).

Reward-model accuracy (RewardBench-style) is a proxy for a proxy! Keep asking what you are *really* optimizing.



THE ALIGNMENT CEILING: OBJECTIVE MISMATCH IN REINFORCEMENT LEARNING FROM HUMAN FEEDBACK

A PREPRINT

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ABSTRACT

Reinforcement learning from human feedback (RLHF) has emerged as a powerful technique to make large language models (LLMs) more capable in complex settings. RLHF proceeds as collecting human preference data, training a reward model on said data, and optimizing a base ML model with respect to said reward for extrinsic evaluation metrics (e.g. MMLU, GSM8k). RLHF relies on many assumptions about how the various pieces fit together, such as a reward model capturing human preferences and an RL optimizer extracting the right signal from a reward model. As the RLHF process involves many distinct design decisions, it is easy to assume that multiple processes are correlated and therefore numerically linked. This apparent correlation is often not true, where reward models are easily overoptimized or RL optimizers can reduce performance on tasks not modeled in the data. Notable manifestations of models trained with imperfect RLHF systems are those that are prone to refusing basic requests for safety reasons or appearing lazy in generations. As chat model evaluation becomes increasingly nuanced, the reliance on a perceived link between reward model training, RL scores, and downstream performance drives these issues, which we describe as an *objective mismatch*. In this paper, we illustrate the causes of this issue, reviewing relevant literature from model-based reinforcement learning, and argue for solutions. By solving objective mismatch in RLHF, the ML models of the future will be more precisely aligned to user instructions for both safety and helpfulness.

1 Introduction

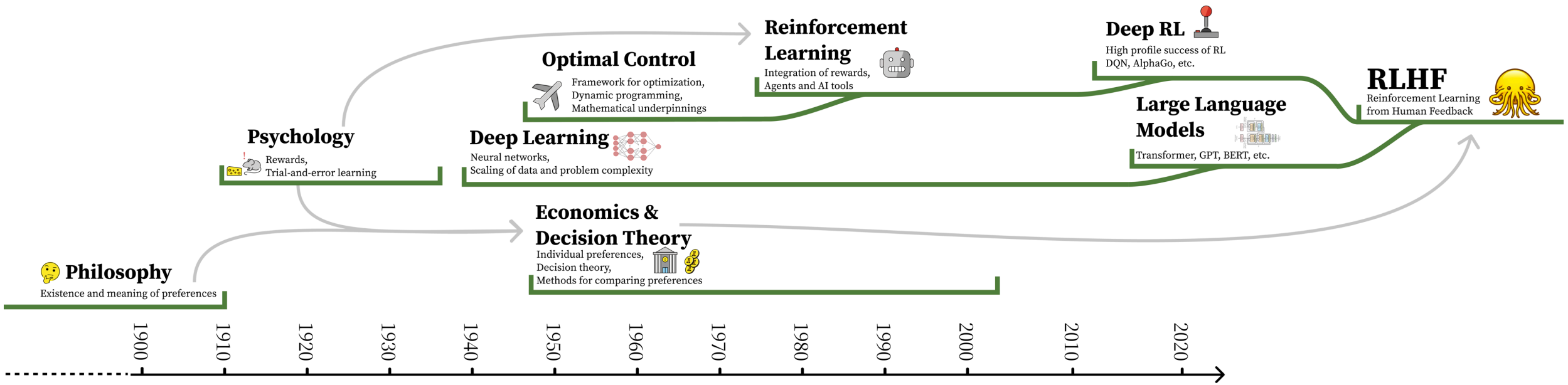
Reinforcement learning from human feedback (RLHF) is a powerful tool for integrating qualitative values into large machine learning models (Bai et al., 2022; Christiano et al., 2017; Ouyang et al., 2022) that are used in popular consumer apps such as ChatGPT and Midjourney. RLHF was popularized with its use to integrate human values into large language models (LLMs) for aligning chat tools (Schulman, Zoph, Kim, & more, 2022; Team et al., 2023). RLHF has become an important technique in the process of making models better at responding to user requests, often referred to as instruction-tuned, steerable, aligned, or chat-tuned.

RLHF methods typically operate in a multi-step process on top of a base language model, first learning a model of human preferences that acts as a reward function, and second using this model within a reinforcement learning (RL) loop. These two steps are often executed independently, with a reward model (RM) being trained on human preference data and then the RL optimizer is used to extract maximum information from the RM into the base model. This multi-step process induces challenges (Schulman, 2023) – even the most popular RLHF models include weaknesses such as `llama-2-70b-chat-hf`'s propensity to refuse vanilla requests on safety grounds (Röttger et al., 2023) or a version of ChatGPT documented officially as having "cases of laziness" (OpenAI, 2024). Colloquially, these issues fall under the potential banner of "too much RLHF." These failures are signs of the current limitations of RLHF, where even with positive signals in training of each individual module, the resulting model can have unintended behaviors.

In this paper, we detail and argue for solving a fundamental challenge in modern RLHF learning schemes – *objective mismatch* – in order to mitigate these issues. In RLHF, three important parts of training are numerically decoupled: the evaluation metrics, the reward model, and the generating model (policy). This mismatch between the reward model and

Part 1: A short history of optimizing or measuring preferences

Many fields converged into “RLHF”



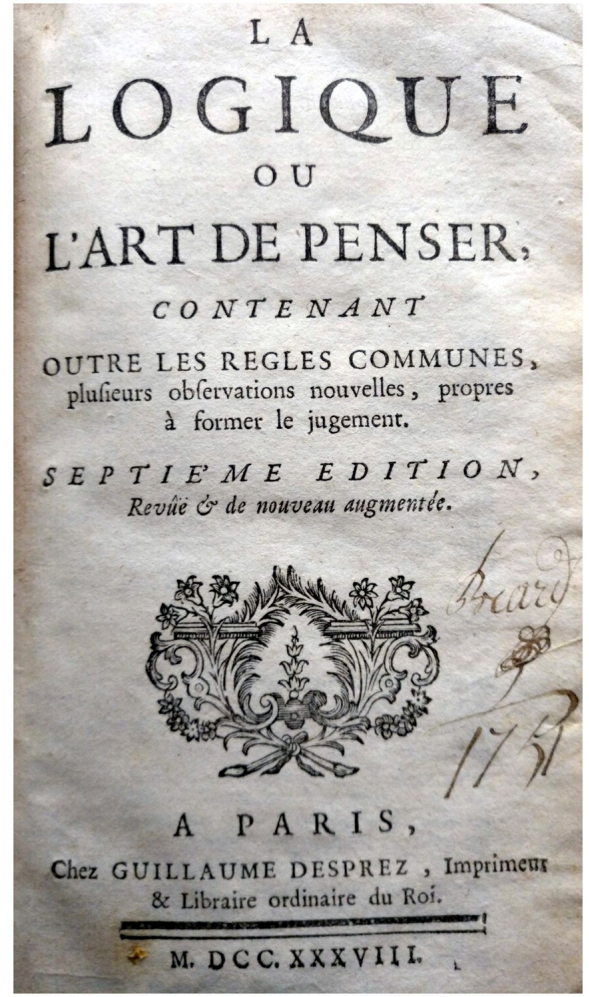
Lambert et al., 2023

Utility, from logic to a number

The idea that choices can be *scored* is old. The common thread: the idea that human wants can be reduced to a single measure:

- **Port Royal Logic** (1662): decision quality = outcome weighted by its probability (Arnauld, 1662)
- **Bentham's hedonic calculus** (early 1800s): weigh all of life on one complicated, but common scale (Bentham, 1823)
- **Ramsey**, *Truth and Probability* (1931): first to quantify preference *and* belief together as the way that individuals make probabilistic decisions (Ramsey, 2016)

> *"To judge what one must do to obtain a good or avoid an evil, it is necessary to consider not only the good and evil in itself, but also the probability that it*


LA
LOGIQUE
OU
L'ART DE PENSER,
CONTENANT
OUTRE LES REGLES COMMUNES,
plusieurs observations nouvelles, propres
à former le jugement.
SEPTIE'ME EDITION,
Revûë & de nouveau augmentée.
A PARIS,
Chez GUILLAUME DESPREZ, Imprimeur
& Libraire ordinaire du Roi.
M. DCC. XXXVIII.

La Logique, ou l'Art de Penser (the Port Royal Logic), 1662

Von Neumann-Morgenstern utility theorem (1947)

Von Neumann & Morgenstern (1947): if your preferences obey a few axioms (completeness, transitivity, continuity, independence), they can be represented by a single **utility function**, and rational choice = maximizing **expected utility** (Von Neumann & Morgenstern, 1947).

This is the result RLHF leans on to justify fitting a scalar reward.

In RLHF, essentially none of those *ifs* hold:

- preferences **drift** during and after labeling
- they're **context- and framing-dependent**
- at high complexity they can be **intransitive**
- and they're **multidimensional**, squashed into one number



John von Neumann (Los Alamos badge photo, public domain)

Bradley-Terry (1952): Comparisons to scores

The statistical model that turns *comparisons* into *scores* – and became the backbone of reward modeling (Bradley & Terry, 1952) (see Chapter 5 / Lecture 2):

$$P(y_w \succ y_l \mid x) = \sigma\big(r(x, y_w) - r(x, y_l)\big) = \frac{e^{r(x,y_w)}}{e^{r(x,y_w)} + e^{r(x,y_l)}}$$

Give it pairwise human comparisons; out comes a scalar reward. This is *why* RLHF needs **preference data** (and where the imperfections enter).

One scalar preference, many features

A reward model compresses, into a single number, all of:

- helpfulness, honesty, harmlessness, tone, format, length, taste…
- the annotator's psychology, culture, and the interface they used
- whatever the *framing* of the comparison nudged them toward

We then optimize hard against that number, which is impossible to ever do perfectly! There will always be trade-offs.

Where utility theory breaks down

Almost as soon as utility was formalized, social choice (the field studying how preferences should be aggregated) and economics found its limits:

- **Arrow's impossibility theorem** (1950): no voting rule aggregates individual preferences into a collective one while satisfying a few basic fairness criteria (Arrow, 1950)
- **Sen**, *Behaviour, Choice and Values* (1973): choice ≠ preference; revealed-preference theory is too thin (Sen, 1973)
- **Hirschman**, *Against Parsimony* (1984): people have *preferences over their preferences* – so preferences may be unmeasurable (Hirschman, 1984)



Kenneth Arrow (CC BY 3.0, via Wikimedia Commons)

Aside: Social choice for AI alignment

If aggregating preferences is the hard part, **social choice theory** is the field that studies it – and a natural lens for alignment.

Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback.

- A 2024 position paper I was part of argues social choice should guide how we aggregate diverse human feedback (Conitzer et al., 2024).
- Whose preferences count? How do you combine disagreeing labelers? ...
- (One of the rabbit holes I used to spend more time in.)



Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback

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Abstract

Foundation models such as GPT-4 are fine-tuned to avoid unsafe or otherwise problematic behavior, such as helping to commit crimes or producing racist text. One approach to fine-tuning, called *reinforcement learning from human feedback*, learns from humans' expressed preferences over multiple outputs. Another approach is *constitutional AI*, in which the input from humans is a list of high-level principles. But how do we deal with potentially diverging input from humans? How can we aggregate the input into consistent data about "collective" preferences or otherwise use it to make collective choices about model behavior? In this paper, we argue that the field of *social choice* is well positioned to address these questions, and we discuss ways forward for this agenda, drawing on discussions in a recent workshop on Social Choice for AI Ethics and Safety held in Berkeley, CA, USA in December 2023.

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1. Introduction

Recently, *reinforcement learning from human feedback* (RLHF) has become the primary strategy that leading AI companies such as OpenAI (OpenAI, 2023), Anthropic (Anthropic, 2023), Meta (Meta, 2023), and Google (Google, 2023) use to make pretrained large language models (LLMs) more capable and controllable (Christiano et al., 2017; Ziegler et al., 2019) and to align them with human values. However, RLHF faces many limitations and challenges (Casper et al., 2023; Lambert & Calandra, 2023), including unrepresentative data (Prabhakaran et al., 2021; Feffer et al., 2023), unrealistic models of human decision-making (Hong et al., 2022; Freedman et al., 2021; Siththaranjan et al., 2023; Lambert et al., 2023), and insufficient modeling of human diversity (Kirk et al., 2023; Freedman et al., 2023) which may lead to political bias (Motoki et al., 2023; Rozado, 2024). We propose that ideas from social choice theory (Arrow, 2012; Fishburn, 1973; Kelly, 1988; Brandt et al., 2015)—e.g., concerning whose preferences should be integrated into decisions and how this should be done—are needed to solve many of these open problems.

While models that are solely pretrained on internet data may produce repetitive or harmful text, RLHF trains models to follow instructions (Ouyang et al., 2022) and produce helpful and "harmless" outputs (Bai et al., 2022a) based on human judgments. RLHF gathers example outputs from a pretrained LLM or examples written by humans. Next, humans are asked to select the outputs that best meet specified criteria (such as being "helpful" or "unbiased"). These judgments, often called *preferences*, are then used to fine-tune the LLM. From a social choice perspective, this method raises several critical questions: Which humans are asked to judge outputs? What criteria do they use? How are their judgments combined? And how do their expressed judgments relate to their actual preferences?

Constitutional AI (CAI), which involves reinforcement learning from AI feedback (RLAIF), is an alternative addressing some of these questions (Bai et al., 2022b). Humans produce a "constitution" that specifies principles that the LLM is trained to align with. However, one must still

1

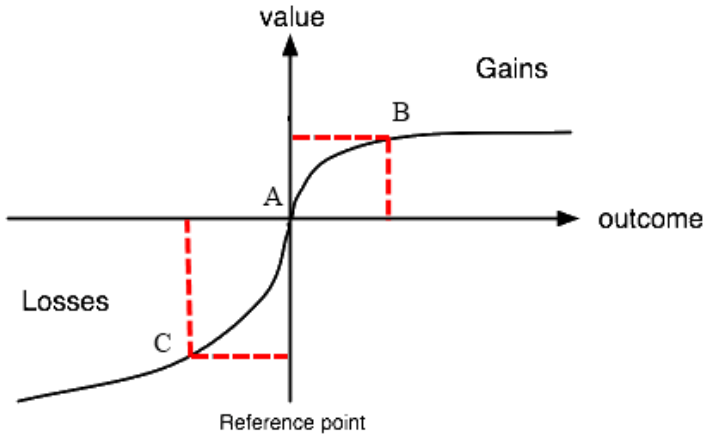
Conitzer et al., 2024

Preferences are not stable objects over time and context

From psychology and behavioral economics:

- Preferences **drift** – they change with time, mood, and experience (Pettigrew, 2019)
- Choices are shaped by situation and framing, not just an inner ranking (Gilbert et al., 2022)

A problem for "collect a label, train on it later repeatedly."



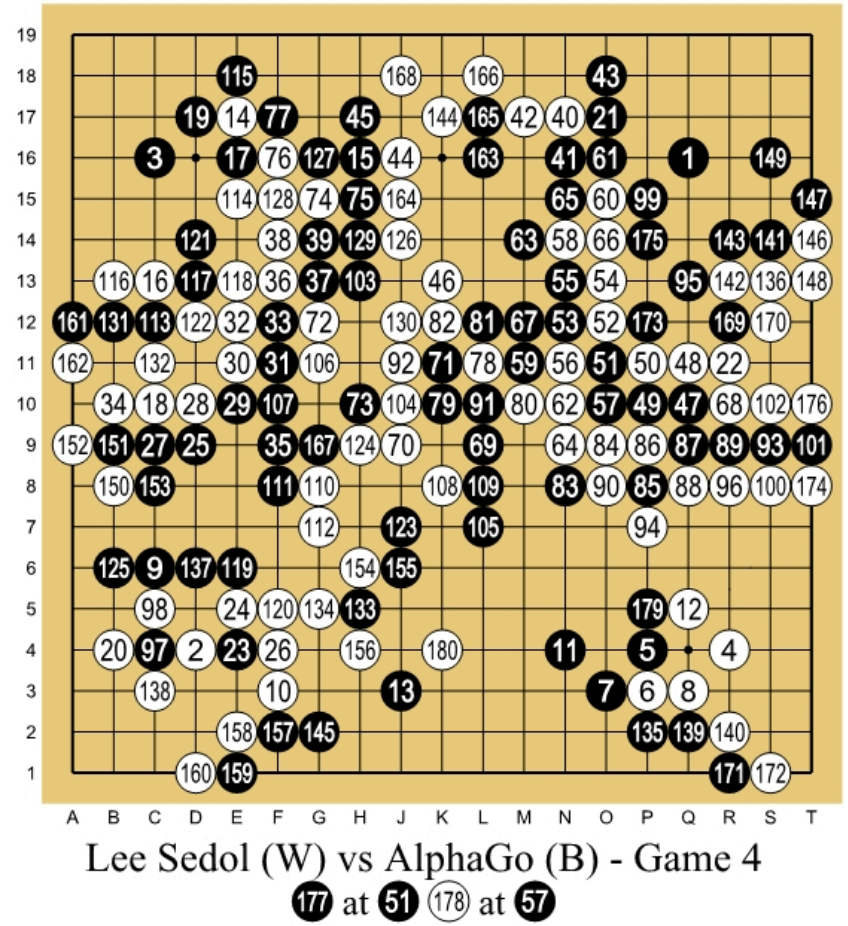
The prospect-theory value function: the same outcome is valued differently depending on the reference point – value depends on framing, not a fixed ranking (CC BY-SA 3.0, via Wikimedia Commons).

The other root: Optimal control & RL

In parallel, a machinery for *optimizing* a reward matured:

- **Bellman** (1957): MDPs and dynamic programming (Bellman, 1957) → origins of "optimal" control / control theory
- **Sutton** (1988): temporal-difference learning for credit assignment (Sutton, 1988) (instead of learning from a lot of past data)
- **Watkins** (1992): Q-learning (Watkins & Dayan, 1992) (learning value)
- **DQN** (2013): deep RL at scale (Mnih et al., 2013); **AlphaGo / AlphaZero** (2017): mastery from self-play (Silver et al., 2017). "RL works"

Note: these guarantees assume a **single, closed-form reward**!



Lee Sedol vs AlphaGo, Game 4 (2016) (CC BY-SA 4.0, via Wikimedia Commons)

Costs ≠ rewards ≠ preferences

These three are **ontologically different**, and the emergence of modern post-training treated them as interchangeable (Lambert et al., 2023).

- **Costs** come from control: physical, measurable, given, and often have clear optimality/bounds.
- **Rewards** are from psychology and an RL convenience: a scalar signal to maximize.
- **Preferences** are human, relational, and unstable – *not* obviously a scalar at all.

Reducing preferences to rewards made the optimization format tractable, but is the root cause of many of the unsolvable "biases" in RLHF and preference data.

Costs ≠ rewards ≠ preferences

These three are **ontologically different**, and the emergence of modern post-training treated them as interchangeable (Lambert et al., 2023). Deep RL's theory lives in MDPs with a fixed, closed-form reward (e.g. games, control).

- A learned reward model is a *moving, noisy proxy*, not a ground-truth reward.
- Inverse RL – learning a reward *from behavior* – is conceptually close but oddly absent from RLHF practice (Ng et al., 2000).
- So we inherit RL's optimizers without inheriting its guarantees.

On what “reward” is

> *Rewards in an RL system correspond to primary rewards, i.e., rewards that in animals have been hard-wired by the evolutionary process due to their relevance to reproductive success. … Further, RL systems that form value functions, … effectively create conditioned or secondary reward processes whereby predictors of primary rewards act as rewards themselves… The result is that the local landscape of a value function gives direction to the system’s preferred behavior: decisions are made to cause transitions to higher-valued states. A close parallel can be drawn between the gradient of a value function and incentive motivation.*

– Singh et al., 2009 (Singh et al., 2009)

Part 2: Preference data – the trade-offs in practice

Where we started – why collect preference data

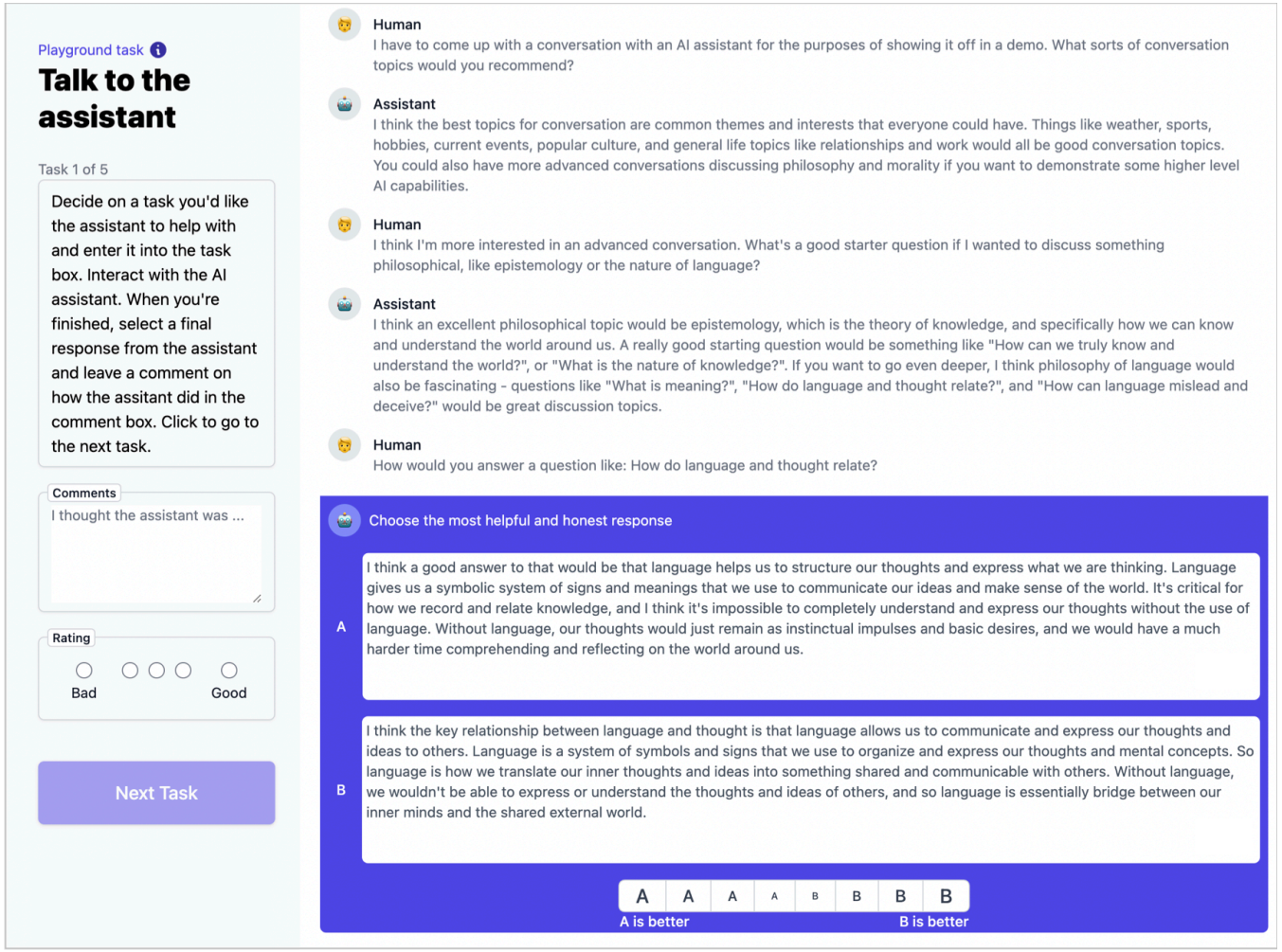
It is far easier to **judge** than to **generate** – humans (and models) can reliably say which of two answers is better long before they can write the better one. There are many cases where models can't at all generate a good answer, but can pick up on cues for which is better.

But collecting (and processing) any data like this well is the most **opaque** part of the modern post-training pipeline.

Very few open models ship fully open human preference data *with* the methodology used to collect it – the best examples are NVIDIA's **HelpSteer** datasets (behind the open Nemotron models) (Wang et al., 2024) (Wang et al., 2024) (Wang et al., 2025).

Example data collection interfaces

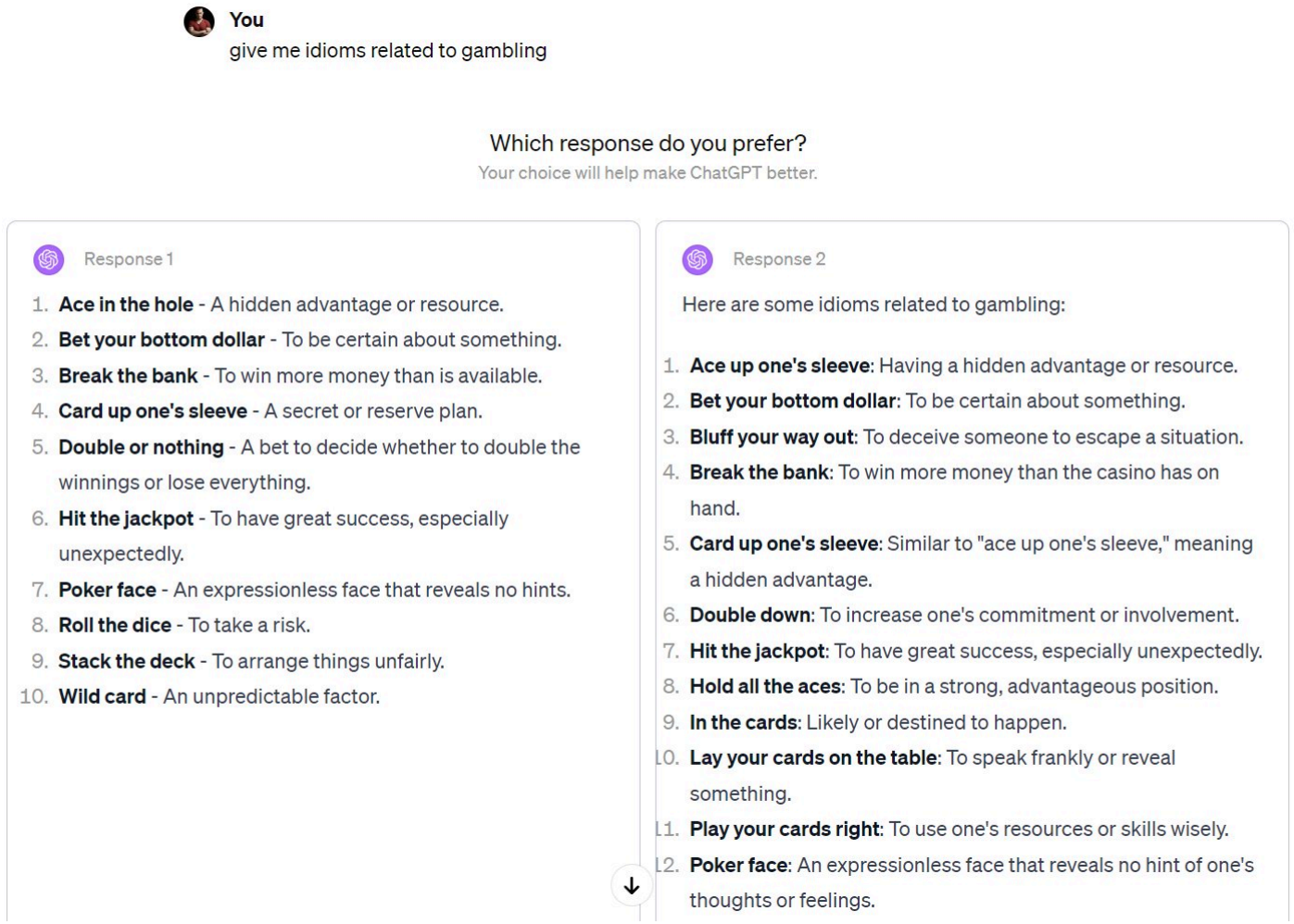
Interface 1: Research data collection (Anthropic's early Claude models)



An early preference-collection interface from Anthropic's research: review the full conversation, then rate. Bai et al., 2022 (CC-BY).

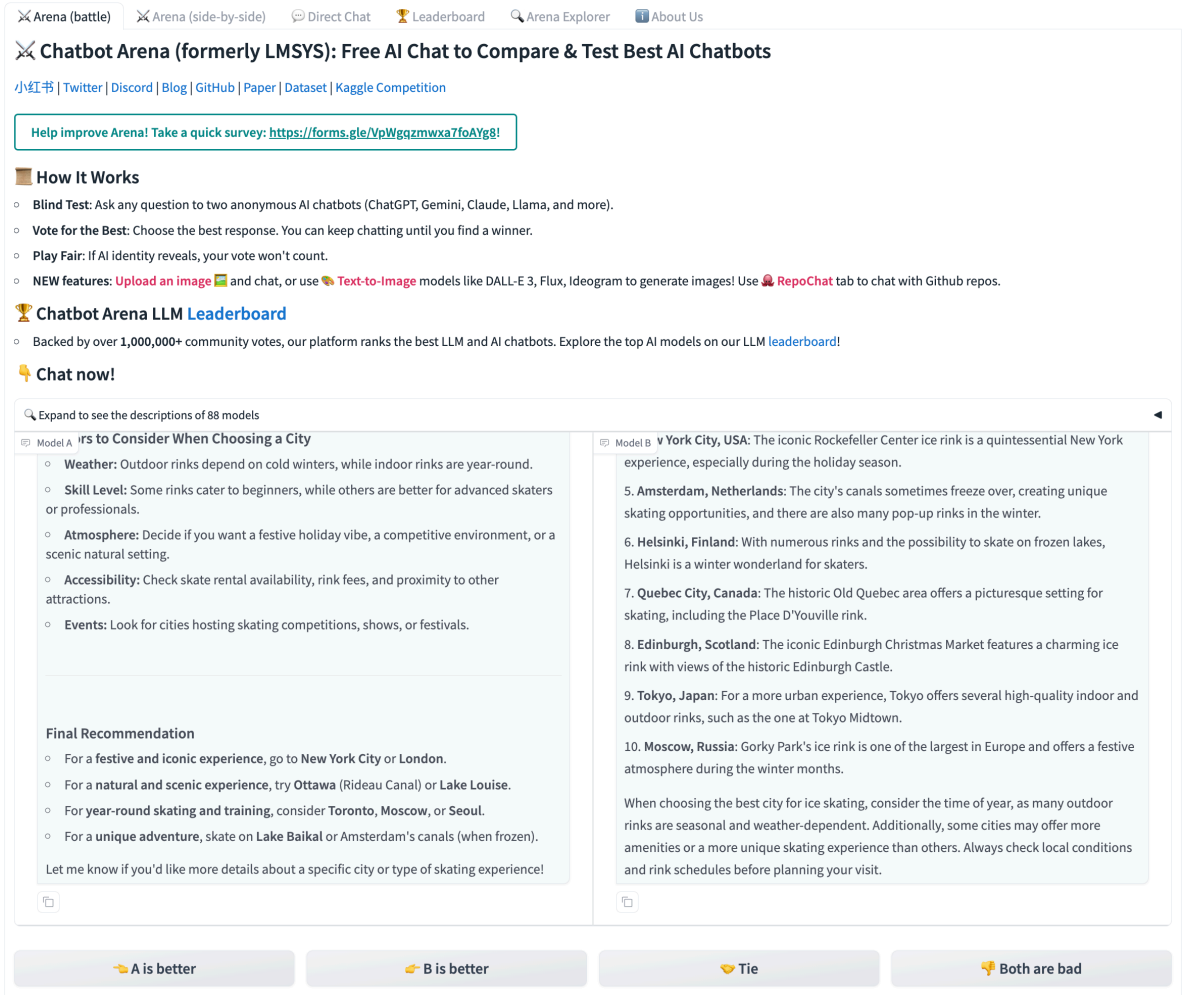
Bai et al., 2022

Interface 2: A/B testing in production (ChatGPT user data)



Two completions from different ChatGPT beta models, served side by side. The answers are very close – a reminder that preference data is noisy and hard to get exactly right.

Interface 3: Pairwise with ties (Chatbot Arena public evaluation)



An early version of the Chatbot Arena interface: pairwise comparison with a tie option.

Chiang et al., 2024

Interface 4: A single bit (Ai2 demos; in many other products)



Up/down voting from Ai2's research demos – the minimal directional signal.

Interface 5: Pick-from-many (default in image models)



Selecting among generated images – preference data outside of text. Every interface shapes the preference it captures.

Rankings vs. ratings

Ratings: a score on one completion in isolation (e.g. 1-5). Good as metadata.

Rankings: relative comparisons, often on a Likert scale – early Claude used an 8-point scale (Bai et al., 2022); UltraFeedback pairs high- vs low-rated completions (Cui et al., 2023).

In practice almost everyone trains on pairwise rankings, binarized to chosen/rejected for the Bradley-Terry loss (lecture 2 / chapter 5) – and keeps ratings on the side.

USER

Rating: "Rate this completion 1-5."

"Paris is the capital of France, known for the Eiffel Tower."

ASSISTANT (Annotator)

4 / 5

USER

Ranking: "Which response is better?"

A: *"Paris."* · **B:** *"Paris – the capital since 987, home of the Eiffel Tower."*

ASSISTANT (Annotator)

B > A

What exactly is a “Likert scale”?

A **Likert scale** records a preference as an *ordered, graded* judgment – not just which answer wins, but **by how much**, on a symmetric scale with an optional neutral middle. (After psychologist Rensis Likert, 1932 – the same “strongly agree … strongly disagree” survey tool, repurposed for pairwise preference.)

5-point with a tie (LMArena-style): labelers can say “about the same.”

1	2	3	4	5
A much better	A better	*tie*	B better	**B much better**

8-point without a tie (early Claude (Bai et al., 2022)): forces a direction.

1	2	3	4	5	6	7	8
A>>>B			A>B	B>A			**B>>>A**

How many points, and whether ties are allowed, are both design choices that change the data you collect.

Structured (synthetic) preference data

In domains with structure, you can build preference pairs automatically:

- **Math**: a correct solution > an incorrect one.
- **Instruction following (IFEval-style)**: prompt twice – with the constraint and without – and prefer the one that obeys it. The constraint is checked with code, so the label is free.

In these narrow domains, structured pairs can beat quality-judged preferences (Lambert et al., 2024). This is a simple form of *synthetic* preference data – chapter 12 / lecture 7.

Example prompt: *"Describe the ocean.* ***Respond in all lowercase.****"*

ASSISTANT (Chosen (sampled with the constraint) ✓)
the ocean covers most of our planet, a restless sheet of salt water…

ASSISTANT (Rejected (sampled without the constraint) ✗)
The ocean covers most of our planet. It is a restless sheet of salt water…

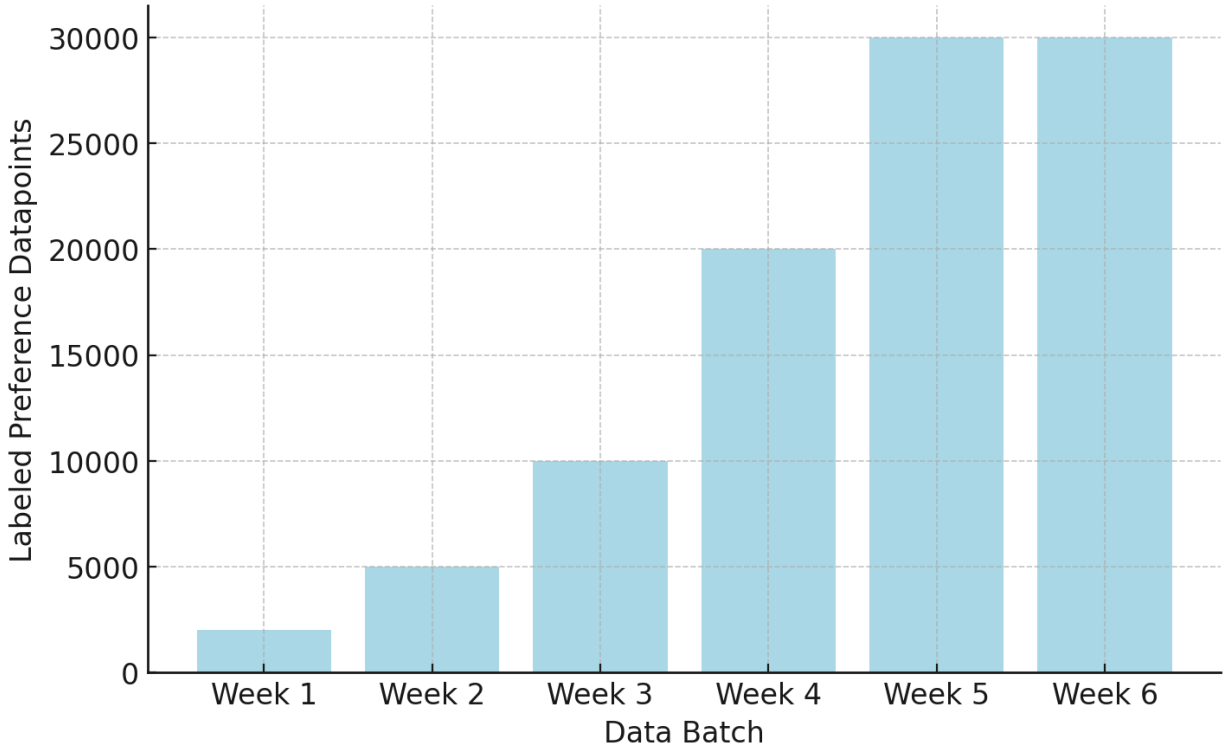
Beyond pairwise preferences

The pairwise comparison is a convention, not a law. Alternatives:

- **Directional / single-bit** labels (thumbs up/down), trained with Kahneman-Tversky Optimization (KTO) (Ethayarajh et al., 2024).
- **Token-level / fine-grained** feedback (Wu et al., 2023): Label specific tokens, or token-spans as good or bad.
- **Natural-language** feedback – written critiques instead of a label (Chen et al., 2024).

Richer signal, harder collection (and not often used extensively in practice).

Sourcing & contracts



A typical multi-batch human-data contract (~$500K): an early ramp where goals and methodology narrow, with much of the first batches thrown out. Larger contracts vary substantially.

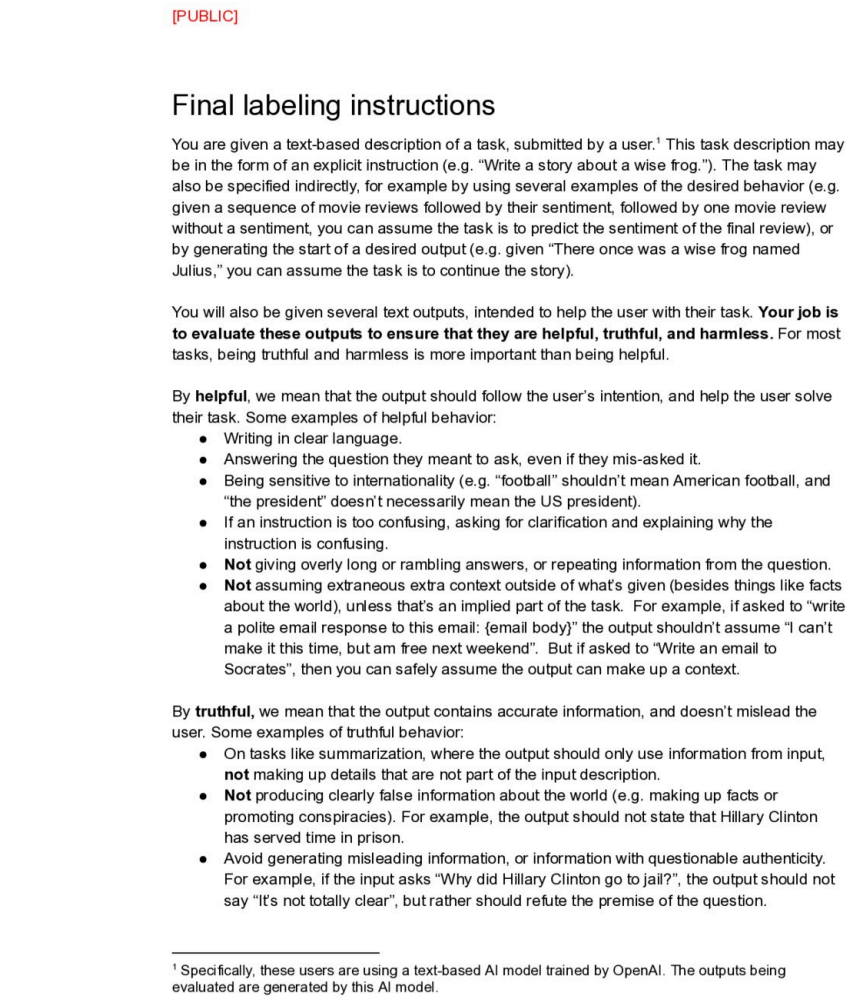
Access is relationship-driven: vendors are supply-limited and favor big budgets and known brands. Millions get spent and partly wasted; few teams have the bandwidth to fully use human data. Contracts often restrict you from releasing some data like this openly.

Is a far more complex industry today with environments, etc. But, even for simple, human preference data, you need a robust, existing post-training recipe to plug it into.

Guiding data collection: Labeler instructions

Once a contract is signed, buyer and vendor agree on **detailed instructions** for every task – normally never seen outside the lab.

- One of the first *public* examples: OpenAI released the full **InstructGPT** labeler instructions (2022) (Ouyang et al., 2022). This was later deleted, but I recovered it.
- Pages of guidance on ranking **helpful, truthful, harmless** – the spec that actually shapes the labels.
- **Download the PDF** (mirrored on rlhfbook.com).



[PUBLIC]

Final labeling instructions

You are given a text-based description of a task, submitted by a user.[1] This task description may be in the form of an explicit instruction (e.g. "Write a story about a wise frog."). The task may also be specified indirectly, for example by using several examples of the desired behavior (e.g. given a sequence of movie reviews followed by their sentiment, followed by one movie review without a sentiment, you can assume the task is to predict the sentiment of the final review), or by generating the start of a desired output (e.g. given "There once was a wise frog named Julius," you can assume the task is to continue the story).

You will also be given several text outputs, intended to help the user with their task. **Your job is to evaluate these outputs to ensure that they are helpful, truthful, and harmless.** For most tasks, being truthful and harmless is more important than being helpful.

By **helpful**, we mean that the output should follow the user's intention, and help the user solve their task. Some examples of helpful behavior:

- Writing in clear language.
- Answering the question they meant to ask, even if they mis-asked it.
- Being sensitive to internationality (e.g. "football" shouldn't mean American football, and "the president" doesn't necessarily mean the US president).
- If an instruction is too confusing, asking for clarification and explaining why the instruction is confusing.
- **Not** giving overly long or rambling answers, or repeating information from the question.
- **Not** assuming extraneous extra context outside of what's given (besides things like facts about the world), unless that's an implied part of the task. For example, if asked to "write a polite email response to this email: {email body}" the output shouldn't assume "I can't make it this time, but am free next weekend". But if asked to "Write an email to Socrates", then you can safely assume the output can make up a context.

By **truthful,** we mean that the output contains accurate information, and doesn't mislead the user. Some examples of truthful behavior:

- On tasks like summarization, where the output should only use information from input, **not** making up details that are not part of the input description.
- **Not** producing clearly false information about the world (e.g. making up facts or promoting conspiracies). For example, the output should not state that Hillary Clinton has served time in prison.
- Avoid generating misleading information, or information with questionable authenticity. For example, if the input asks "Why did Hillary Clinton go to jail?", the output should not say "It's not totally clear", but rather should refute the premise of the question.

[1] Specifically, these users are using a text-based AI model trained by OpenAI. The outputs being evaluated are generated by this AI model.

InstructGPT "Final labeling instructions" (OpenAI, 2022) – released publicly.

Ouyang et al., 2022

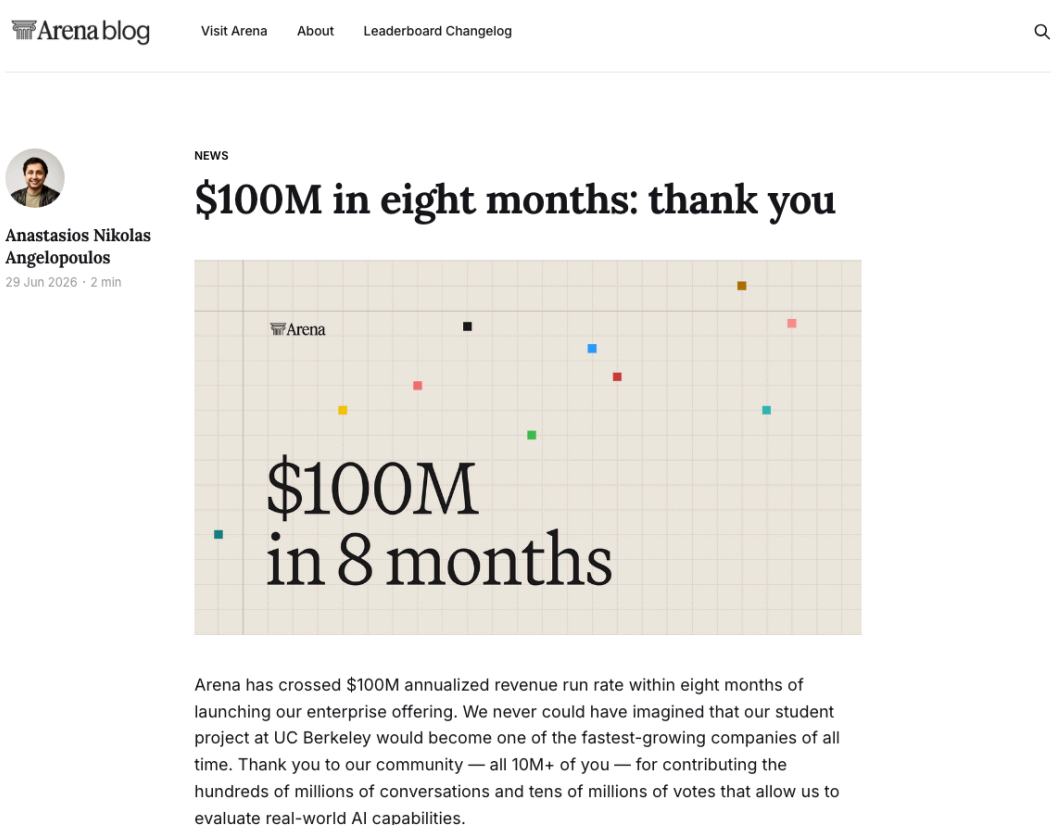
A dataset we bought: No Robots

On Hugging Face's **H4 team**, we commissioned human data the same way the labs did.

- **No Robots** (Rajani et al., 2023) – 10K expert human-written demonstrations, matching InstructGPT distribution, paid for from a vendor and then released **openly** (rare for commissioned data): hf.co/datasets/HuggingFaceH4/no_robots
- From the era/team of the **Zephyr** models (Tunstall et al., 2024) and the **Open LLM Leaderboard** (Beeching et al., 2023).

Preference *evals* are now a standalone business

Recently in 2026 **Arena** (formerly, the LMArena leaderboard, formerly Chatbot Arena) reached a **~$100M annualized revenue run-rate within ~8 months** of launching its enterprise A/B testing offering (Arena, 2026).



Arena, 2026

Part 3: Open questions in preference data

Bias: What to watch for in data

Subtle, systematic biases sail straight from the data into the model:

- **Prefix bias** – the opening disproportionately drives the label (Kumar et al., 2025)
- **Sycophancy** – agreeing with the user over being right (Sharma et al., 2024)
- **Verbosity** – longer rated higher (Singhal et al., 2023)
- **Formatting** – lists and bold look "better" (Zhang et al., 2025)
- **Flattery / fluff** – decorative language inflates scores (Bharadwaj et al., 2025)

Detecting and controlling these biases is central to collecting high-quality preference data – and they are exactly what over-optimization amplifies into model behavior (Lecture 9).

Higher-level complexities

- **Collection context** – do workplace labels transfer to end users? Paid vs. volunteer? Do annotators follow instructions or their own values?
- **Type of feedback** – does a binary pair actually capture the preference we mean? What structure mirrors how people really compare?
- **Population & demographics** – who labels? Is disagreement **noise or signal**?
- **Are the preferences even expressed in the models?**

The unaudited gap: Spec → data → behavior

RLHF's *motivation* (align to human preference) has drifted from its *practice* (make models effective).

Because industrial RLHF is closed, we can't check whether the trained model actually reflects the spec given to annotators. The **Model Spec** (OpenAI, 2024) documents intended behavior, but the link from data → behavior stays largely unaudited.

Many of these questions already surfaced with synthetic data (chapter 12 / lecture 7): the human/AI feedback balance, and on-policy preference data.

The nature of preferences is the lasting problem of RLHF

This is one of the least-settled, most human parts of the field. Read widely and go to the primary sources. Is a great academic problem!

The course so far

0. Prerequisites review
1. Overview *(ch. 1–3)*
2. IFT, reward models & rejection sampling *(ch. 4, 5, 9)*
3. RL: Motivation & math *(ch. 6)*
4. RL: Implementation & practice *(ch. 6)*
5. The rise of reasoning models *(ch. 7)*
6. Direct preference optimization *(ch. 8)*
7. Synthetic data & modern post-training *(ch. 12)*
8. **Preferences & preference data** *(ch. 10–11) – today*
9. **Over-optimization & RLHF's bad reputation** *(ch. 14, app. B) – next*
10. Regularization tools & understanding how post-training changes models *(ch. 15)*

Thank you

Questions / discussion

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rlhfbook.com



References (1/5)

Arena. "*Arena: The AI Leaderboard Everyone Uses Is Now a |$$100M Business*." 2026.

Arnauld, A. "*The Port-Royal Logic*." 1662.

Arrow, K. "*A difficulty in the concept of social welfare*." *Journal of political economy*, 1950.

Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., et al. "*Training a helpful and harmless assistant with reinforcement learning from human feedback*." *arXiv preprint arXiv:2204.05862*, 2022.

Beeching, E., Fourrier, C., Habib, N., Han, S., Lambert, N., et al. "*Open LLM Leaderboard*." *Hugging Face*, 2023.

Bellman, R. "*A Markovian decision process*." *Journal of mathematics and mechanics*, 1957.

Bentham, J. "*An Introduction to the Principles of Morals and Legislation*." 1823.

Bharadwaj, A., Malaviya, C., Joshi, N., and Yatskar, M. "*Flattery, Fluff, and Fog: Diagnosing and Mitigating Idiosyncratic Biases in Preference Models*." 2025. [link]

Bradley, R., and Terry, M. "*Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons*." *Biometrika*, 1952.

Chen, A., Scheurer, J., Campos, J., Korbak, T., Chan, J., et al. "*Learning from Natural Language Feedback*." *Transactions on Machine Learning Research*, 2024.

References (2/5)

Chiang, W., Zheng, L., Sheng, Y., Angelopoulos, A., Li, T., et al. "*Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference*." *International Conference on Machine Learning (ICML)*, 2024.

Conitzer, V., Freedman, R., Heitzig, J., Holliday, W., Jacobs, B., et al. "*Position: Social Choice Should Guide AI Alignment in Dealing with Diverse Human Feedback*." *International Conference on Machine Learning (ICML)*, 2024. [link]

Cui, G., Yuan, L., Ding, N., Yao, G., Zhu, W., et al. "*Ultrafeedback: Boosting language models with high-quality feedback*." 2023.

Ethayarajh, K., Xu, W., Muennighoff, N., Jurafsky, D., and Kiela, D. "*Kto: Model alignment as prospect theoretic optimization*." *arXiv preprint arXiv:2402.01306*, 2024.

Gilbert, T., Dean, S., Zick, T., and Lambert, N. "*Choices, Risks, and Reward Reports: Charting Public Policy for Reinforcement Learning Systems*." *arXiv preprint arXiv:2202.05716*, 2022.

Hirschman, A. "*Against parsimony: Three easy ways of complicating some categories of economic discourse*." *Bulletin of the American Academy of arts and Sciences*, 1984.

Kumar, A., He, Y., Markosyan, A., Chern, B., and Arrieta-Ibarra, I. "*Detecting Prefix Bias in LLM-based Reward Models*." *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2025.

Lambert, N., and Calandra, R. "*The Alignment Ceiling: Objective Mismatch in Reinforcement Learning from Human Feedback*." 2023. [link]

References (3/5)

Lambert, N., Amos, B., Yadan, O., and Calandra, R. "*Objective Mismatch in Model-based Reinforcement Learning*." *Learning for Dynamics and Control (L4DC)*, 2020. [link]

Lambert, N., Gilbert, T., and Zick, T. "*Entangled preferences: The history and risks of reinforcement learning and human feedback*." *arXiv preprint arXiv:2310.13595*, 2023.

Lambert, N., Morrison, J., Pyatkin, V., Huang, S., Ivison, H., et al. "*Tülu 3: Pushing Frontiers in Open Language Model Post-Training*." *arXiv preprint arXiv:2411.15124*, 2024.

Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., et al. "*Playing atari with deep reinforcement learning*." *arXiv preprint arXiv:1312.5602*, 2013.

Ng, A., Russell, S., and others. "*Algorithms for inverse reinforcement learning.*." *Proceedings of the Seventeenth International Conference on Machine Learning*, 2000.

OpenAI. "*Introducing the Model Spec*." 2024. [link]

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., et al. "*Training language models to follow instructions with human feedback*." *Advances in Neural Information Processing Systems*, 2022.

Pettigrew, R. "*Choosing for changing selves*." *Oxford University Press*, 2019.

Rajani, N., Tunstall, L., Beeching, E., Lambert, N., Rush, A., et al. "*No Robots*." *Hugging Face repository*, 2023.

References (4/5)

Ramsey, F. "*Truth and probability*." *Readings in Formal Epistemology: Sourcebook*, 2016.

Sen, A. "*Behaviour and the Concept of Preference*." *Economica*, 1973.

Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askell, A., et al. "*Towards Understanding Sycophancy in Language Models*." *The Twelfth International Conference on Learning Representations*, 2024. [link]

Silver, D., Schrittwieser, J., Simonyan, K., Antonoglou, I., Huang, A., et al. "*Mastering the game of go without human knowledge*." *Nature*, 2017.

Singh, S., Lewis, R., and Barto, A. "*Where do rewards come from*." *Proceedings of the annual conference of the cognitive science society*, 2009.

Singhal, P., Goyal, T., Xu, J., and Durrett, G. "*A long way to go: Investigating length correlations in rlhf*." *arXiv preprint arXiv:2310.03716*, 2023.

Sutton, R. "*Learning to predict by the methods of temporal differences*." *Machine learning*, 1988.

Tunstall, L., Beeching, E., Lambert, N., Rajani, N., Rasul, K., et al. "*Zephyr: Direct Distillation of LM Alignment*." *First Conference on Language Modeling*, 2024.

Von Neumann, J., and Morgenstern, O. "*Theory of games and economic behavior, 2nd rev.*" *Princeton university press*, 1947.

References (5/5)

Wang, Z., Dong, Y., Delalleau, O., Zeng, J., Shen, G., et al. "*HelpSteer2: Open-source dataset for training top-performing reward models*." *arXiv preprint arXiv:2406.08673*, 2024.

Wang, Z., Dong, Y., Delalleau, O., Zeng, J., Shen, G., et al. "*HelpSteer2-Preference: Complementing Ratings with Preferences*." *arXiv preprint arXiv:2410.01257*, 2024.

Wang, Z., Zeng, J., Delalleau, O., Shin, H., Soares, F., et al. "*HelpSteer3-Preference: Open Human-Annotated Preference Data across Diverse Tasks and Languages*." *arXiv preprint arXiv:2505.11475*, 2025.

Watkins, C., and Dayan, P. "*Q-learning*." *Machine learning*, 1992.

Wu, Z., Hu, Y., Shi, W., Dziri, N., Suhr, A., et al. "*Fine-grained human feedback gives better rewards for language model training*." *Advances in Neural Information Processing Systems*, 2023.

Zhang, X., Xiong, W., Chen, L., Zhou, T., Huang, H., et al. "*From lists to emojis: How format bias affects model alignment*." *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.